

SUMMARY

Software Engineer passionate about building scalable, data-driven applications and robust system architectures. Proficient in modern cloud technologies, databases, and infrastructure automation. Motivated to contribute to innovative 0–1 initiatives at the intersection of engineering and science.

EDUCATION

Northeastern University

September 2023 – May 2025

Master of Science in Software Engineering Systems

Relevant Courses: Concepts of Object Oriented Design using Java, Web Design/User Experience Design, Program Structures and Algorithms, Design Patterns

St. Francis College for Women

June 2018 – June 2021

Bachelor of Science in Honors (Software Engineering)

Relevant Courses: OOP through Java, Data Structures, GUI through Java, Object Oriented Software Engineering, Web Programming

TECHNICAL SKILLS

Programming Languages: Java 8+, JavaScript, TypeScript, Python, (familiar with Go)

Backend Development: RESTful APIs, Node.js, Express, Spring Boot, scalable microservices

Frameworks & Libraries: React.js, JavaFX, Socket.io

Databases: PostgreSQL, MySQL, MongoDB, SQL Server

Cloud & Infrastructure: AWS (EC2, RDS, Route 53, VPC, IAM), Terraform, CI/CD (GitHub Actions)

Software Engineering: Data structures, design patterns, distributed systems, test-driven development (TDD), Agile/Scrum

WORK EXPERIENCE

Elorce Technologies

April – August 2022

Quality Analyst Intern

- Designed and executed 50+ comprehensive test cases for backend APIs and UI components, increasing test coverage by 40% and boosting system reliability by 30%.
- Optimized API response times by 20% by identifying and resolving 15+ priority backend bugs in collaboration with developers.
- Conducted root cause analysis for production-level issues, contributing to a 98% transaction success rate and customer satisfaction.

PROJECTS

Modular Java CLI Food Ordering System

April 2025

- Developed a modular CLI-based food ordering application with 12+ features including order placement, payment processing, and cloning functionality, improving scalability by 60%.
- Implemented 10+ design patterns (Command, Strategy, Observer, Facade, Builder) to reduce code duplication by 35% and increase maintainability.
- Optimized CSV-based data storage and object handling, boosting performance by 90% under high-volume transaction loads.

Cloud-Native Scalable User Management Platform

December 2024

- Built a scalable backend with secure authentication and automated email verification via AWS Lambda, SQS, and Node.js, reducing manual onboarding time by 80%.
- Utilized AWS RDS, VPC, IAM roles, and Route 53 for infrastructure setup using Terraform, achieving 99% uptime.
- Designed and implemented CI/CD pipelines using GitHub Actions, cutting deployment errors by 60% and enabling auto-scaling on load-balanced EC2 instances.

JavaFX Recipe Desktop Application

December 2023

- Built a feature-rich Java application with user login, content creation, and dynamic recipe browsing, increasing usability by 30%.
- Applied OOP principles and Java design patterns to emulate RESTful service logic and support scalable local CRUD operations for 1,000+ daily user requests.
- Integrated MongoDB for persistent data storage, achieving 99.9% data accuracy and seamless migrations.

Full Stack Music Playlist App

December 2023

- Engineered a secure full-stack music application with JWT & Bcrypt-based authentication, supporting 1,000+ concurrent sessions and ensuring data privacy.
- Integrated Socket.io and CORS to reduce interaction latency by 30%, improving real-time user engagement.
- Developed RESTful APIs using Node.js and Express to manage playlist CRUD operations, enabling seamless communication.